Evolutionary and homolog comparison

Comparison of CDS:

```
XM_003874496.1      ------------------------------------------------------------        0
PMM2_CDS_WT         ------------------------------------------------------------        0
XM_072942246.1      AGTTCCGGGCCGAGTTCCAGGTGCCAGCGAGTCCTGTAAGGTACGCGTGGAAGTTGGGGT        60


XM_003874496.1      -------ATGGGCTCCAAGGCTATTCTTCTCTTTGACGTTGATGGCACCCTCACGCCGCC        53
PMM2_CDS_WT         -ATGGCAGCGCCTGGCCCAGCGCTCTGCCTCTTCGACGTGGATGGGACCCTCACCGCCCC        59
XM_072942246.1      TATGGCGGCACCCAGGCCAGCGCTCTGCCTCTTCGATGTAGACGGGACCCTGACGGCCCC        120
                                  **  *    ***** ** ** ** ** ***** **  * **

XM_003874496.1      GCGCAACCCGGAAACCCATGACATGAAGGAGGCGCTGTTGAAGGCTCGTGCAGCTGGGTT        113
PMM2_CDS_WT         GCGGCAGAAAATTACCAAAGAAATGGATGACTT---CCTACAAAAATTGAGGCAGAAGAT        116
XM_072942246.1      GCGGCAGAAAATTACCAAAGACATGGATGGCTT---TCTACAAGAACTGAGGCAGAAGAT        177
                    ***  *       *** * ** *** * *         *  *              * *

XM_003874496.1      CAAGCTTGGCGTCGTCGGTGGCTCCGACTTTGCCAAGCAGAAGGAGCAGCTGGGCGAGTC        173
PMM2_CDS_WT         CAAAATCGGAGTGGTAGGCGGATCGGACTTTGAGAAAGTGCAGGAGCAACTGGGAAATGA        176
XM_072942246.1      CAAAATTGGAGTAGTAGGCGGGTCGGACTTTGAGAAAGTACAGGAGCAGCTGGGAAGTGA        237
                    ***  * ** ** ** ** ** ** *******  **     ******* *****

XM_003874496.1      -----------------------------------GATTCTGGAGGACTTTGACTATGT        197
PMM2_CDS_WT         -----------------------------------TGTGGTTGAAAAATACGATTATGT        200
XM_072942246.1      TGGTAAACTACACCTTGCTCGTTACATTTTGGTAAAAGTGGTTGAAAAATATGATTATGT        297
                                                      *  * **  * *  ** *****

XM_003874496.1      GTTTTCCGAGAACGGTCTATTGGCCTACAAGGATGGCAAAGAGTTTCACCGCAACAGCTT        257
PMM2_CDS_WT         GTTTCCAGAAAATGGCTTGGTAGCATACAAAGATGGGAAACTCTTGTGTAGACAGAATAT        260
XM_072942246.1      GTTTCCAGAAAATGGCTTGGTAGCATACAAAGATGGGAAACTCTTATGCAAACAGAATAT        357
                    **** * ** ** **  *  * ** ***** ***** ***   **       * *   *

XM_003874496.1      GCTGAGGGCTCTTGGGAACGAGAAGGTTGTGGCATTTGTAAAGAAGTGTCTGCACCTCAT        317
PMM2_CDS_WT         TCAAAGTCATCTGGGTGAGGCCCTAATCCAAGATTTAATCAACTACTGTCTGAGCTACAT        320
XM_072942246.1      TCAGGGTCACCTGGGCGAGGCCCTGATCCAAGATTTAATCAACTACTGTCTGAGCTACAT        417
                     *   *    ** **  * *     *    *  **  * **  * ******  *  ***
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In the image above and in the following page we can see the partial comparison of the CDS files of the PMM2 protein of Leishmania mexicana, Homo sapiens and Vicugna Pacos.

The spaces below the three sequences correspond to differences in the three sequences, the asterisks correspond to parts of the sequences that are all the same nucleotide.

There are quite a bit of conserved regions. In general there aren't long sequences of non-coinciding DNA, with the exception of the edges and a certain sequence present in Vicugna Pacos but not in the other two.

In the beginning and ending the sequence differences are more frequent, which makes sense from an evolutionary point of view, as edges are more prone to mutate.

Then there's the GAP caused by the stretch (nucleotides 238 to 272 in Vicugna sequence) in the Vicugna CDS not present in the other two DNA sequences.

In addition, there's a few stretches in which the Leishmania sequence presents consistent substitute variants in respect to the other two (ex nucleotides 87 to 109 in Leishmania sequence).

```
XM_003874496.1      CGCCGATTTGGACATTCCAGTGCAGCGCGGCACGTTTGTGGAGTTCCGCAACGGTATGTT       377
PMM2_CDS_WT         TGCGAAAATTAAACTCCCGAAGAAGAGGGGTACTTTCATTGAATTCCGAAATGGGATGTT       380
XM_072942246.1      TGCGAAAATTAAACTCCCGAGGAAGAGGGGTACTTTCATTGAATTTCGAAACGGGATGTT       477
                     **  *  *  *  * **   * ** * ** ** **  * ** ** ** ** ** *****

XM_003874496.1      CAACGTATCTCCCATCGGCCGCAACTGCTCGCAGCAGGAGCGTGATGAGTTCGAGAATCT       437
PMM2_CDS_WT         AAACGTGTCCCCTATTGGAAGAAGCTGCAGCCAAGAAGAACGCATTGAGTTCTACGAACT       440
XM_072942246.1      GAATGTGTCCCCCGTTGGACGAAGCTGCAGCCAGGAAGAACGCATTGAGTTCTACGAACT       537
                     ** ** ** **  * **  * * ****   **  * ** **   ******* *  * **

XM_003874496.1      CGACAAGGAGCGCCACATCCGTGAGAAGCTCATCAGGGAGCTCAAGGAGGCCTTCCCTGA       497
PMM2_CDS_WT         CGATAAAAAAGAAAATATAAGACAAAAGTTTGTAGCAGATCTACGGAAAGAGTTTGCTGG       500
XM_072942246.1      TGATCAAAAAGAAAACATAAGGCAGAAATTCGTGGAGGATCTGCGGAAGGAGTTTGCAGG       597
                     **  *  *    * **  *  * **  *  *   ** **   * * *  **  * *

XM_003874496.1      CTACCAGCTTGCCTATTCCGTAGGTGGTCAAATCAGTTTTGACGTCTTCCCAAAGGGGTG       557
PMM2_CDS_WT         AAAAGGCCTCACGTTTTCCATAGGAGGCCAGATCAGCTTTGATGTCTTTCCTGATGGATG       560
XM_072942246.1      AAAAGGCCTCACGTTCTCCATAGGAGGCCAGATCAGCTTTGATGTCTTCCCTGACGGATG       657
                      *    **  * *  *** **** ** ** ***** ***** ***** **  * ** **

XM_003874496.1      GGATAAGACTTACTGCCTTCAGTTTGTTGAGAATGACTTTGAAACCA---TTCACTTTTT       614
PMM2_CDS_WT         GGACAAGAGATACTGTCTGCGACATGTGGAAAATGACGGTTATAAGACCATTTATTTCTT       620
XM_072942246.1      GGACAAGCGGTACTGCCTGAGACACGTGGAAAACGATGGCTATAAGACCATTTATTTCTT       717
                    *** ***   ***** **      ** ** ** **    * *  *   ** * ** **

XM_003874496.1      CGGTGATAAGACGTCAGAGGGCGGCAACGATTATGAGATCTTCACTGATTCTCGCACAAT       674
PMM2_CDS_WT         TGGAGACAAAACTATGCCAGGTGGCAATGACCATGAGATCTTCACAGACCCCAGAACCAT       680
XM_072942246.1      TGGAGACAAAACCATGCCTATATGAGGAGACGCAAGACTTTGGGCTCATAAAATCTTCTG       777
                     ** ** ** **          *    **       * *   *  *

XM_003874496.1      CGGTCATTCTGTGAAGACATATA--AGGATACAATTGCGATTCTTGAGGCACTTCTCGA-       731
PMM2_CDS_WT         GGGCTACTCCGTGACAGCGCCTG-----AGGACA-----CGCGCAGGATCTGTGAACTGC       730
XM_072942246.1      AAAAT-ATCTGTCTGAAGACCTGTTCTGCCAGTTTTCTCAGAGCAGAGAGACCTCATTCC       836
                           ** **        *                     *

XM_003874496.1      ------------------GGATTCGCGGTGA----------------------------       744
PMM2_CDS_WT         TGTTCTCCTAA-------------------------------------------------       741
XM_072942246.1      TGATCTCCATCCTGGACTCCTTTCAAGGTGTGTTGAAGGTCAGTGGCTGCAGCAGCTCAT       896


XM_003874496.1      ------------------------------------------------------------       744
PMM2_CDS_WT         ------------------------------------------------------------       741
XM_072942246.1      GATTTAATCATTGTACAGGTAGGCGGCAAGTGCCAGGGTGATTGCCTCTAAAGCGTTAAG       956


XM_003874496.1      ------------------------------------------------------------       744
PMM2_CDS_WT         ------------------------------------------------------------       741
XM_072942246.1      ATAGTGAGTGCCAGTGCTTATGAACCCCTTTGGTCATAAATTTGATCGTGGCTGGGGAAG       101
```

The rest of the comparison is the Vicugna Pacos CDS alone, which is 5795 nucleotides long.

Comparison of protein sequence:

However, the gene isn't necessarily representative as the variations of codons could code for the same or similar amino acid. To further analyze we compare protein sequences.

```
XP_003874545.1      --MGSKAILLFDVDGTLTPPRNPETHDMKEALLKARAAGFKLGVVGGSDFAKQKEQLGES      58
PMM2_protein_WT     MAAPGPALCLFDVDGTLTAPRQKITKEMDDFLQK-LRQKIKIGVVGGSDFEKVQEQLGND      59
XP_072798347.1      MAAPRPALCLFDVDGTLTAPRQKITKDMDGFLQE-LRQKIKIGVVGGSDFEKVQEQLGSD      59
                          *: ********* **:  *::*.  * :     :*:******** * :****..

XP_003874545.1      ------------ILEDFDYVFSENGLLAYKDGKEFHRNSLLRALGNEKVVAFVKKCLHLI      106
PMM2_protein_WT     ------------VVEKYDYVFPENGLVAYKDGKLLCRQNIQSHLGEALIQDLINYCLSYI      107
XP_072798347.1      GKLHLARYILVKVVEKYDYVFPENGLVAYKDGKLLCKQNIQGHLGEALIQDLINYCLSYI      119
                                ::*.:**** ****:****** : ::.:   **:  :  ::: **  *

XP_003874545.1      ADLDIPVQRGTFVEFRNGMFNVSPIGRNCSQQERDEFENLDKERHIREKLIRELKEAFPD      166
PMM2_protein_WT     AKIKLPKKRGTFIEFRNGMLNVSPIGRSCSQEERIEFYELDKKENIRQKFVADLRKEFAG      167
XP_072798347.1      AKIKLPRKRGTFIEFRNGMLNVSPVGRSCSQEERIEFYELDQKENIRQKFVEDLRKEFAG      179
                    *.:.:* :****:******:****:**.***:** ** :**::.:**:*:: :*:: * .

XP_003874545.1      YQLAYSVGGQISFDVFPKGWDKTYCLQFVEND-FETIHFFGDKTSEGGNDYEIFTDSRTI      225
PMM2_protein_WT     KGLTFSIGGQISFDVFPDGWDKRYCLRHVENDGYKTIYFFGDKTMPGGNDHEIFTDPRTM      227
XP_072798347.1      KGLTFSIGGQISFDVFPDGWDKRYCLRHVENDGYKTIYFFGDKTMPI------------      226
                      *::*:**********.**** ***:.**** ::**:******

XP_003874545.1      GHSVKTYKDTIAILEALLEDSR     247
PMM2_protein_WT     GYSVTAPEDTRRICELLFS---     246
XP_072798347.1      ----------------------     226
```

In the image above we can see the comparison of the protein sequences of the PMM2 protein of Leishmania mexicana, Homo sapiens and Vicugna Pacos.

Here asterisks indicate that all three sequences coincide in aminoacid, the colons indicate that the amino acids don't coincide but have similar physicochemical properties. The dots and spaces indicate more important variances.

The proteins are similar in length but we see, similarly to the CDS comparison, the variants concentrate especially towards the edges of the proteins, which in this case includes the Vicugna protein sequence ending much earlier than the other two.

There's also a gap caused by a part of the sequence of the Vicugna protein not present in the other two, which coincides with the same gap at the nucleotide level.

At the same time, there seems to be two stretches which are marked as low conservation specifically because the Leishmania has different amino acids than the other two species (substitute variant). This also could be observed at the gene level.

The mutation present in the mutant PMM2 associated with PMM2-GCD is in a region that doesn't show a lot of conservation.

Comparison of protein secondary structure:

Finally, to see how the mutations could affect the shape and formation of the proteins, we compare the secondary structure of the Leishmania Mexicana PMM2 with the Human Wild Type PMM2.

Below we can see the full protein sequences for both species, each amino acid accompanied by the secondary structure that it belongs to (provided by professors).

This structure is by far the most conserved, the secondary structure is almost completely identical for the two sequences.

There's only one stretch in which the secondary structure seems to not be conserved, and it doesn't coincide with the one from the primary structure. This one goes from 65th to 84th protein (counting with Human sequence as reference).

The mutation present in the mutant PMM2 associated with PMM2-GCD is in the middle of the last alpha helix of the sequences.

